

Generation Interconnection System Impact Study Report

For Queue Project Q14-0341

Newark 138 kV

May 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0341. The Interconnection Customer proposes a 229.1 MW wind facility interconnecting at the Newark 138 kV substation in the Pennsylvania area. The request was received on June 10, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 224 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,774,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Primm - Johns Crk 138kV is overloaded to 156.2% of its 188 MVA emergency rating for the loss of the Lions Rd Substation - Piney Creek Project 500kV. The project contributes 22.2% DFAX (50.8 MW) to this overload.

The Knott - Primm Substation 138kV is overloaded to 136.9% of its 184 MVA emergency rating for the loss of the Kirkland Park - Allegheny Hydro No 9 138kV. The project contributes 29.4% DFAX (67.3 MW) to this overload.

For the loss of Piney Creek Project - Lions Road 500kV, the Phoenix Hydro Project - Newark 138kV loads to 108.0% of its 225 MVA rating. Project DFAX: 38.6%.

Elizabethton - Johns Crk 138kV: 105.3% loading (188 MVA) following the outage of the Newark - Phoenix Hydro Project 138kV; DFAX = 16.9%.

For the loss of Newark - Phoenix Hydro Project 138kV, the Elizabethton - Tesoro Kenai Cogeneration Plant 138kV loads to 103.1% of its 196 MVA rating. Project DFAX: 16.9%.

For the loss of Knott - Primm 138kV, the Allegheny Hydro No 9 - Kirkland Park 138kV loads to 101.9% of its 237 MVA rating. Project DFAX: 25.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Primm - Johns Crk 138kV	Rebuild line with higher-capacity conductor	\$6,053,000	71% (\$4,306,000)
Knott - Primm Substation 138kV	Rebuild line with higher-capacity conductor	\$2,176,000	70% (\$1,517,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Newark - Phoenix Hydro Project 138kV Line	Reconductor line	\$1,904,000	90% (\$1,707,000)
Elizabethton - Johns Crk 138kV	Reconductor line	\$865,000	67% (\$576,000)
Elizabethton - Tesoro Kenai Cogeneration Plant 138kV	Reconductor line	\$1,724,000	67% (\$1,148,000)
Allegheny Hydro No 9 - Kirkland Park 138kV	Reconductor line	\$1,975,000	77% (\$1,520,000)
	Total allocated to project		\$10,774,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.